Locations and Binding Domains*

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Abstract

This paper argues that some noncomplementary instances of anaphors and pronouns in the context of spatial prepositions conceal different structures underlying fixed spatial relations (places) and dynamic ones (paths). I propose that place prepositions form opaque binding domains due to the necessary mediation of a functional head *p*which sets out a clause-like structure, while paths are transparent to binding unless they contain an overt place preposition.

Keywords: *Binding theory, Anaphors, Logophoricity, Modern Hebrew, Spatial prepositions.*

1 Introduction

Many languages have different classes of pronouns for short- and long-distance coreference (Lees and Klima 1963; Langacker 1966; Faltz 1977; Chomsky 1981; Reinhart 1983; Heim 1998; Büring 2005; among many others). This well-known pattern is exemplified below by the English complex anaphor *themselves* and the pronoun *them*.

(1) a. Some people$_1$ like {them$_{*1}$/themselves$_1$}.

b. Some people$_1$ think their children$_2$ like {them$_{1/*2}$/themselves$_{*1/2}$} .

The choice between the anaphor and the pronoun in these sentences is clearly correlated with the distance of the antecedent. The anaphor corefers with the most local DP that c-commands it,

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while the pronoun is separated from its antecedent by an embedded subject (*their children*). This regularity often breaks under prepositions of spatial relations, as seen in some classic examples in (2).

(2) a. John$_1$ saw a snake next to {him$_1$/himself$_1$}.

b. Max$_1$ rolled the carpet over {him$_1$/himself$_1$}. (Reinhart and Reuland 1993: 67)

Here, either a pronoun or an anaphor is acceptable with antecedents at the same distance. Understanding such sentences require of English speakers to learn in which phrases the pronoun and the anaphor may refer to the same entity, a non-trivial task since not all prepositions allow the same freedom. For instance, in (3), the pronoun and the anaphor are complementary in reference despite the context of the spatial preposition *through*.

(3) Superman can see through {him$_{*1}$/himself$_1$}.

This variation has led previous accounts of spatial anaphors to abandon the premise of a distance-based analysis in favor of broad contextual factors like hierarchies between referents (Kuno 1987; Ariel 2001), speakers' expectations (Haspelmath 2008; Ariel 2008), or fine-grained spatial nuances (Wechsler 1997; Lederer 2013). I argue that this conclusion is premature and that pronouns and anaphors follow distance-based rules in spatial PPs as they do elsewhere, including cases in which their distribution seems to overlap.

I will propose a solution based on a syntactic difference between PPs that overtly encode a fixed location and those that do not. Specifically, I will argue that prepositions of fixed location introduce a local subject within the prepositional phrase, while prepositions of dynamic location lack this property. Assuming standard definitions of syntactic locality, it will follow that prepositions of fixed location form binding domains that enable pronouns to corefer with higher DPs without violating Principle B. Prepositions of dynamic locations are generally transparent to binding, meaning an anaphor is required for coreference. Overlaps between pronouns and anaphors are expected in two cases: contexts that license exempt anaphors, and prepositions that can be read as either fixed or dynamic. The data presented below confirms through independent diagnostics that controlling for these exceptions resolves spatial PPs toward the pronoun or the anaphor in a predictable fashion.

1.1 Background

There is some independent evidence that prepositions are not syntactically uniform, particularly when comparing spatial prepositions and the rest of the category (Jackendoff 1973; Hoekstra 1988; Reinhart and Reuland 1993; Rooryck 1996; Rooryck and Vanden Wyngaerd 2007; Botwinik-Rotem 2008; den Dikken 2010b). Prepositions like *of*, *for*, and the nonspatial versions of *to* and *from* are predetermined by the predicates they combine with, as shown below in (4a), which means they have no clear effect on the meaning of the sentence. By contrast, spatial PPs come with a broad selection of heads (4b), each leading to different truth conditions.

(4) a. ready {for/*to/*from/*of}; divorce {*for/*to/from/*of}; add {*for/to/*from/*of}
   b. walk {to/toward/around/under} the bridge.

Prepositions of the former type are also blocked from stacking on one another (5a), which is an option for spatial prepositions (5b).

(5) a. *talk [for [to [the children]]].
   b. arrive [from [under [the bridge]]].

Déchaine (????) treats prepositions as a borderline category on the functional-lexical axis and lists further differences in properties such as valency and C-selection. The recurring observation is that some prepositions are richer in meaning and structure than others, and spatial prepositions tend to be on the higher end of this scale. Jackendoff (1973) ties this status to a multilayered syntax that separates fixed areas in space and dynamic shifts between areas, known by the respective terms Place and Path.[1]. Most prepositions are intuitively linked with one class or the other, as illustrated for English in Table 1.

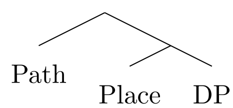
Fixed relations in space (Places)	**Dynamic relations in space (Paths)**
in, on, next to, in front of, behind	*to, toward, through, over, from*

Table 1: Spatial prepositions in English

[1]The notion of areas has received two separate lines of formalization: as regions (**?**) or as vectors in space (Zwarts, 1997; **?**). The current discussion will not bear on this question. I also abstract away from more fine-grained divisions of the path notion (Jackendoff, 1973; Lederer, 2013)

We will see shortly that this categorization can be misleading, but for now let us assume that each preposition realizes one of these two primitive concepts.[2]. The same opposition is often described by the respective terms Location and Direction, though the term "Directional" may be used collectively for PPs that denote change of location regardless of whether the preposition itself denotes a path or a place, a point that will become relevant in Section 3.3. Jackendoff's position is that the notion of path contains a fixed place, and that this is reflected in a universal structure in the form of (6).

(6)



This syntax served as the basis for further projections capturing various semantic nuances of spatial relations that will play a limited role here (Koopman 2000; Svenonius 2006; Rooryck and Vanden Wyngaerd 2007; den Dikken 2010b). The crucial point for the current purposes is that spatial notions are built as trees of rising complexity. Another component of the proposal is that parts of this tree may be covert, and it has been widely adopted that change of location invariably follows from a path over place structure, even if some projections are not pronounced. That is, a path phrase like *from the house* includes a covert place with a default meaning akin to the preposition *at*, associating the PP in (7a) with the underlying structure in (7b). Similarly, a silent path with the meaning *to* explains change of location in (8a) (Koopman, 1994, 2000).[3].

(7) a. They came from the house.

b. [PATH from [PLACE (at) [DP the house]]].

(8) a. They jumped in the lake.

b. [PATH (to) [PLACE in [DP the lake]]].

This connects to a broader question on the syntax of complex notions. Later work by Jackendoff argues that the syntax mirrors the conceptual structure, which means paths have to contain place projections since they are dynamic relations over fixed spatial areas (Jackendoff, 1983). I will

[2] The prepositions *to* and *from* are also used for the abstract goal/source relation that is not anchored to a point in space (e.g., *talk to x*, *hear from x*). I will specifically refer to the spatial reading of these prepositions unless otherwise stated.

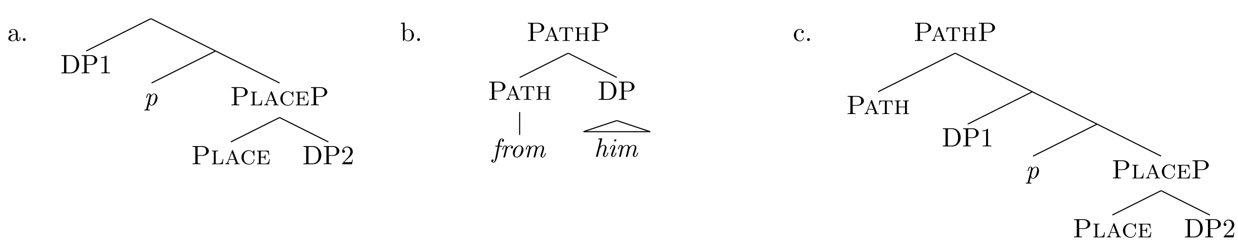
[3] Koopman analyzes silent *to* as a trace of a path that incorporates with a motion verb. See den Dikken (2010b) and Gehrke (2008) for alternatives

therefore pose two related questions in this paper: (i) Can an independently motivated syntax of spatial PPs explain the relevant binding facts?, and (ii) does this syntax have to match the PPs conceptual structure?

1.2 Main claim: Place and path have a different syntax

Setting out from the Path-Place distinction, I will first show that the preference for a pronoun over an anaphor in PPs is tightly correlated with having an overt place preposition. Path PPs that lack this projection will turn out to behave like direct objects and require an anaphor to corefer with the subject. From this I will conclude that place prepositions form opaque binding domains by introducing a PP-internal subject, and that this step is skipped in path phrases that lack an overt place component. The idea that PPs have subjects is well-precedented in literature going back to Talmy (1978) and Hoekstra (1988), who describe spatial prepositions as subject-object relations between an entity and a spatial reference (Figure and Ground in Talmi's terms). It has became a standard analysis for spatial PPs (Kayne 1984; den Dikken 2006; Svenonius 2003; Ramchand and Svenonius 2004; Mateu and Acedo-Matellán 2012,a.o), which effectively assigns them a small clause structure. The data reviewed in the upcoming section suggests that this configuration is compatible with PPs containing a place preposition, and that not all spatial PPs contain a place, at least in the syntactic sense.

I will propose the following set of structures for change of location instead of a uniform path-over-place model. A Place PP (9a) has both an internal and an external argument, which I assume is mediated by a functional project akin to little v in verbs (following Svenonius 2003). A bare path (9b) has a single argument, while an overt path-over-place phrase contains an embedded two-place predication (9c).

(9)


Out of these options, (9a) and (9c) are rather standard options in the current view of spatial prepositions. I will argue that the observed patterns of coreference support (9b) for PPs that lack an overt place projection. I will base on data from English as the main object language of previous accounts and introduce Hebrew data when there is a need to control for exempt anaphors. I predict that the current conclusions will extend to other languages that express spatial relations through prepositions, with some variation due to language-specific structural differences.

In the rest of this paper, Section 2 presents some technical details on how I distinguish path from place (Section 2.1) and control for so-called exempt anaphors (Section 2.2). The main empirical part in Section 2.3 establishes a generalization regarding the relevance of overt place prepositions to binding. Section 3 presents my proposal for why only overt place arguments form binding domains and discusses some exceptions. Section 4 concludes the paper.

2 Place PPs are binding domains

2.1 Differentiating path and place

I adopt the traditional classification by which place PPs appear with stative predicates like (10a), while Path PPs require a motion event at least implicitly (10b) (Jackendoff, 1973).[4] Table 4 lists the acceptability values of English spatial prepositions in these contexts.

(10) a. The girls stayed {in/on/next to/behind/beside/under/around/*to/*toward/*onto/ *into/*through} the car.

b. The girls skipped {in/on/next to/behind/beside/under/around/to/toward/onto/ into/through} the car.

behind	*in front of*	*against*	*between*	*next to*	*off*	*around*	*at*	*past*	*upon*	*among*
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
across	*beside*	*along*	*into*	*onto*	*to*	*toward*	*from*	*through*		
✓	✓	*	*	*	*	*	*	*		

Table 2: English prepositions' compatibility with stative verbs

This test effectively diagnoses fixed place readings but does not rule out dynamic ones. That is, PPs that are acceptable with statives may still denote change of location in other contexts. This is

[4]Path prepositions may nonetheless occur in stative contexts if they are inside nouns that imply some kind of motion, as in *a road through the mountains.* I rely on stative verbs as a diagnostics.

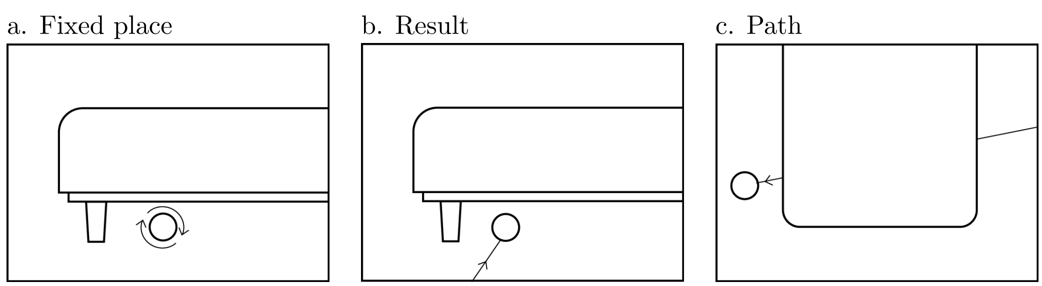
common for a preposition like *in*, seen below denoting a stative location in (11a) and a dynamic one in (11b).

(11) a. The geese stayed in the pond.

b. The geese jumped in the pond.

The context in (11b) has two known readings: one comparable to (11a), in which the geese stay in the pond while jumping, and one where jumping moves the geese into the pond from a location outside of it. The standard view is that *in* remains a fixed place prepositions in both cases, but its import in the dynamic reading is the location of the subject at the end of a motion event (Hoekstra 1988; Rothstein 2006; Gehrke 2008; den Dikken 2010b). I will show in Section 2.3 that these prepositions indeed behave like fixed places for the purposes of binding, and yet that there are also contexts in which some place prepositions can be understood as literal paths (Section 2.3.3). Such prepositions may give rise to three distinct readings, illustrated below with *under*.

(12) The ball rolled under the bed.



I will distinguish readings (a-b) from (c) using a test adapted from Ramchand 2008 of coordinating a stative sentence that refers to an object's final location, like *it stayed there*. This will only be compatible with PPs that describe a fixed place, in this case the first two scenarios.

(13) The ball rolled under the bed and stayed there. [✓Fixed place, ✓Final place, #Path]

I will therefore use stative predicates as a two-step diagnostics: first to sort prepositions by their basic meaning, and then to rule out parallel path readings of place prepositions.

2.2 Excluding logophors

A major confound for a distance-based analysis of spatial anaphors is the extended availability of antecedents that are construed as the source of information, which may bypass intervening subjects (Ross 1970; Cantrall 1974; Pollard and Sag 1992; Reinhart and Reuland 1993; Charnavel and Sportiche 2016). Below are some famous instances of this phenomenon, known as logophoricity (due to Sells 1987) or exempt anaphors (Pollard and Sag, 1992).

(14) a. Max$_1$ boasted that the queen invited Lucie and himself$_1$ (/him$_1$) for tea. (Reinhart and Reuland 1993: 26a)

b. The paper was written by Ann and myself(/me). (Ross 1970: 21a)

c. John$_1$ was going to get even with Mary. That picture of himself$_1$(/him$_1$) in the post office would really annoy her. (Pollard and Sag 1992: 46)

The antecedents in (14) appear in a different clause than the anaphor (14a), in a completely different sentence (14c), or are altogether missing in the given context (14b). Importantly, all are naturally construed as the source that delivers the information in the utterance, or the center of perspective from which it is uttered. The following examples show that tweaking these contexts such that the antecedent is not the center makes the anaphors unacceptable.

(15) a. *I boasted that the queen invited Lucie and himself for tea.

b. *Have you seen Joe? The paper was written by Ann and himself.

c. *Mary has had it with John. That picture of himself in the post office really annoyed her.

A common view of this contrast associates perspective centers with syntactic reflexes at various points along the sentence, which then stand as local antecedents for anaphors (Svenonius 2006; Rooryck and Vanden Wyngaerd 2007; Charnavel 2019; Charnavel and Sportiche 2016). A particularly detailed account in Charnavel and Sportiche 2016 assumes operators that set the perspective locally and project covert logophoric pronouns, which in turn mediate what appears superficially to be long-distance anaphors. In this view, a large portion of the cases of free choice between pronouns and anaphors are actually a choice between a neutral perspective and the perspective of the antecedent.

Following this line of analysis provides a distance-base explanation for some of the overlapping licensing of pronouns and anaphors in spatial PPs and elsewhere. Since any asserted sentence has a source that is a sentient being, it could trigger the use of anaphors in what would otherwise be a pronoun position. However, logophoricity does not explain the lack of choice in PPs like those below in (16), which I adapted from attested web examples. Here, the anaphors cannot be switched with pronouns and keep the same reference regardless of perspective.

(16) a. The officer$_1$ aimed the gun at himself$_1$ (/*him$_1$).

b. Kobe Bryant$_1$ likes to pass the ball to himself$_1$ (/*him$_1$) off the backboard.

c. Try to copy the folder$_1$ into itself$_1$ (/*it$_1$).

d. Can Superman$_1$ see through himself$_1$ (/*him$_1$)?

My working assumption in this study is that contrasts that are unaccounted for by logophoricity are due to differences in the PPs' syntax. I rely on theories of logophoricity (mainly Charnavel and Sportiche 2016) in order to control for perspective effects in the data under consideration. Accordingly, I will consider two coreferential positions as local to each other when the lower position bans pronouns that obey Principle B of the Binding Theory (English *her/him/it*), or allows anaphors that cannot be used logophorically, such as English *itself* and Hebrew *acm-x*. Pronouns are prevented from local coreference regardless of perspective, which means that when coreference fails between an antecedent and a pronoun, all else being equal, they are part of the same binding domain. The English inanimate anaphor serves as another locality diagnostics as its antecedent is by definition not a center of perspective. The Hebrew anaphor does a similar job since it categorically fails in logophoric usage despite being otherwise parallel to English *x-self* (Bassel 2018; Angelopoulos and Bassel 2019), as seen below in the Hebrew translations of (14).

(17) Modern Hebrew (compare with 14):

a. max$_1$ samax še-ha-malka hizmina et lusi ve-{oto$_1$/ *et acm-o$_1$} le-te
Max rejoiced that-DET-queen invited ACC Lucie and-him ACC REFL-3SG.M for-tea.
'Max rejoiced that the queen invited Lucie and *him/himself for tea.'

b. ha-ma'mar ha-ze nixctav al.jedej an ve-{al.jadi/ *al-jedej acm-i}
DET-paper DET-this written by Ann and-by.me by REFL-1SG
'This paper was written by Ann and me/*myself.'

c. ha-tmuna šel-{o$_1$/ *acm-o$_1$} ba-do'ar hifti'a et ron$_1$.
DET-picture of-3SG.M REFL-3SG.M at.DET-post surprised ACC Ron
'The picture of him/*himself at the post office surprised Ron.'

Section 2.3 will show that absent logophoric mediation, the choice between pronouns and anaphors aligns with the place-path distinction.

2.3 Coreference over prepositions

We have seen that PPs (i) have different levels of complexity and (ii) show diversity in the licensing of pronominal elements. My goal is to connect these factors such that the choice of pronominal element for any given PP follows from its syntactic complexity in accordance with the standard distance-based view.

2.3.1 Pronouns

As a starting point, consider a PP that overtly contains a path and a place layer.

(18) Bryant$_1$ grabs the ball [PATH from [PLACE behind [DP him$_1$]]].

The pronoun inside the PP can be coreferenced with *Bryant*, as we would have expected from an embedded clause like that in (19) (repeated from 1). That is, from a binding perspective this PP has the complexity of a sentence.

(19) People$_1$ think [TP their children$_1$ like them$_1$].

What component of the PP is responsible for its clausal status? The following versions of (18) show that the pronoun retains the coreferential meaning when the path projection is dropped overtly (20), or altogether (21). However, dropping the place projection makes coreference unavailable (22), as with direct objects (23).

(20) Bryant$_1$ kicks the ball [PATH [PLACE behind [DP him$_1$]]].

(21) Bryant$_1$ keeps the ball [PLACE behind [DP him$_1$]].

(22) Bryant$_1$ grabs the ball [PATH from [DP him$_{*1}$]].

(23) Bryant$_1$ grabs [DP him$_{*1}$].

The judgments on the first and last examples are predicted in standard theory: the PP in (20) has the same meaning components as the one in (18) and should therefore be of a similar syntactic complexity, while subject-object coreference is by definition local (23). A surprising contrast turns up between the two middle examples: The PP that consists of a place without a path is acceptable with a coreferenced pronoun, but a path without a place blocks this reading. The view that every path contains a place (conceptually and in the syntax) makes a false prediction here, that the PP in (22) would contain the one in (21) along with its clause-like status.

A possible solution is to analyze *from* in (22) as an oblique case marker rather than a path projection, which would make it syntactically similar to (23). Yet the following sentences show that the ban on coreferenced pronouns is not unique to *to* and *from*: unambiguous path prepositions also trigger Principle B violations (24), while place PPs regularly circumvent it (25).

(24) *Path prepositions*:

Bryant$_1$ throws the ball {toward/through} him$_{*1}$.

(25) *Place prepositions*:

Bryant$_1$ throws the ball {over/under/next to/behind/in front of} him$_1$.

A straightforward explanation for this consistency is that path and place prepositions do not form equal domains for pronouns: place phrases have a similar effect to clause boundaries, while paths by themselves are part of the verb phrase for binding purposes. The bottom line is that PPs may express change of location using a place preposition (26a), a path preposition (26b), or both (26c), but only those with place prepositions form binding domains that protect pronouns from Principle B violations (26a,c).

(26) *Coreference in change of location:*

a. Bryant$_1$ drops the ball behind him$_1$.

b. Bryant$_1$ grabs the ball from him$_{*1}$.

c. Bryant$_1$ grabs the ball from behind him$_1$.

Importantly, all three cases instantiate *him* as a reference point of a spatial relation. That is, they contain a fixed location at the conceptual level, but this does entail a syntactic effect unless there is a pronounced place preposition. I will henceforth refer to this phenomenon as the overt place effect, as per the definition in (27).

(27) Overt Place Effect: A prepositional phrase constitutes an opaque binding domain if it contains an overt place projection.

The following subsection shows that the same effect appears in anaphors when controlling for perspective.

2.3.2 Anaphors

If (27) is the correct generalization, anaphors should show a complementary pattern and appear only in path phrases that lack a place projection. This predicts that, out of the examples in (24-25) above, only the paths will allow an anaphor coreferring with *Bryant*. However, logophoric licensing is expected to completely obscure this contrast by allowing anaphors across the board, as indeed happens in (28).

(28) Bryant throws the ball {to/toward/over/above/under/around/next to/behind/ in front of} himself.

Now consider comparable contexts in which the antecedents are inanimate, replacing the thrown ball with an electric current or radio waves.

(29) *Path prepositions*:

The sensor$_1$ sends a 100mA current to/toward/through{*it$_1$/itself$_1$}.

(30) *Place prepositions*:

a. The radar$_1$ sends electromagnetic pulses {next to/in front of/behind}{it$_1$/*itself$_1$}.

b. The radar$_1$ sends electromagnetic pulses {under/over/around}{it$_1$/itself$_1$}.

(31) *Path+Place*:

The radar$_1$ picks up signals from under {it$_1$/*itself$_1$}.

This fits the prediction in the sense that the PPs with an overt place phrase all license a coreferential pronoun (30 and 31), while those without it require an anaphor (29). What is unpredicted is the subset of place prepositions that allow coreference with either pronominal term (30b), which cannot be dismissed on account of being logophoric now that the antecedent is inanimate. Admittedly, machines and other objects can be personified, in which case their "perspective" might be centered, but this should equally apply to the prepositions in (30a). The following examples show that the same pattern emerges with Hebrew anaphors, which tend to be banned from logophoric licensing regardless of animacy (as we have seen in 17).

(32) Hebrew, Path prepositions:

a. brajent$_1$ zorek et ha-kadur el- {*av$_1$/ acm-o$_1$}.
Bryant throws ACC DET-ball to- 3SG.M REFL-3SG.M
'Bryant throws the ball to *him/himself.'

b. brajent$_1$ zorek et ha-kadur le'ev(e)r{-*o$_1$/ acm-o$_1$}.
Bryant throws ACC DET-ball toward-3SG.M REFL-3SG.M
'Bryant throws the ball toward *him/himself.'

c. ha-flaŝ$_1$ zorek et ha-kadur {*dark-o$_1$/ derex acm-o$_1$}.
DET-Flash throws ACC DET-ball through-3SG.M through REFL-3SG.M
'The Flash throws the ball through *him/himself.'

(33) Place prepositions:[5,6]

a. brajent$_1$ zorek et ha-kadur lejad{-o$_1$/ *acm-o$_1$}.
Bryant throws ACC DET-ball next.to-3SG.M REFL-3SG.M
'Bryant throws the ball next to him/*himself.'

b. brajent$_1$ zorek et ha-kadur mul{-o$_1$/ *acm-o$_1$}.
Bryant throws ACC DET-ball against-3SG.M REFL-3SG.M
'Bryant throws the ball in front of him/*himself.'

c. brajent$_1$ zorek et ha-kadurim mis(a)viv {-o$_1$/ le-acm-o$_1$}.
Bryant throws ACC DET-balls around -3SG.M to-REFL-3SG.M
'Bryant throws the balls around him/himself.'

d. brajent$_1$ zorek et ha-kadur me'al {-av$_1$/ acm-o$_1$}.
Bryant throws ACC DET-ball over -3SG.M REFL-3SG.M
'Bryant throws the ball over him/himself.'

e. brajent$_1$ zorek et ha-kadur mitax(a)t {-av$_1$/ le-acm-o$_1$}.
Bryant throws ACC DET-ball under -3SG.M to-REFL-3SG.M
'Bryant throws the ball <u>under</u> him/himself.'

[5] The object *kadur* 'ball' is pluralized in (33c) to set up an appropriate context for *sviv-o* 'around him'.
[6] Sentences (33c,33e) feature a semantically vacuous *le-* with the anaphor. A more formal register will also feature *le-* with the pronoun form *-o*.

To restate the problem: we have now seen two types of anaphors that are not licensed as logophors and yet appear parallel to pronouns with certain prepositions. The following section proposes that prepositions that enable such overlaps are ambiguous between a path and a place reading. Specifically, I will argue that prepositions like *around*, *under*, *over*, and their Hebrew parallels can be interpreted as a path head in addition to their canonical place reading, and that these readings are the context that require an anaphor for coreference.

2.3.3 Overlaps

The goal in this section is showing that there is both syntactic and semantic evidence that certain prepositions are ambiguous between place and path readings, and so may appear respectively with either a pronoun or an anaphor. I will focus entirely on Hebrew to eliminate the confounding logophoric usage of English animate anaphors. Consider again the contexts where pronouns and anaphors overlap, which I repeat in (34).

(34) a. brajent$_1$ zorek et ha-kadur me'al-{ av$_1$/ acm-o$_1$}.
Bryant throws ACC DET-ball over- 3SG.M REFL-3SG.M
'Bryant throws the ball over him/himself.'

b. brajent$_1$ zorek et ha-kadur mitax(a)t-{ av$_1$/ le-acm-o$_1$}.
Bryant throws ACC DET-ball under 3SG.M to-REFL-3SG.M
'Bryant throws the ball under him/himself.'

c. brajent$_1$ zorek et ha-kadurim mis(a)viv-{ o$_1$/ le-acm-o$_1$}.
Bryant throws ACC DET-balls around 3SG.M to-REFL-3SG.M
'Bryant throws the balls around him/himself.'

The prepositions in these sentences pass the basic test for fixed places, as seen below, which explains how coreferential pronouns are licensed. I argue anaphors are licensed in the same position because these prepositions can also be read as paths.

(35) ha-kadurim niš'aru {me'al/ mitaxat/ misaviv} la-sal.
DET-balls stayed over/ under/ around DET-basket
'The balls stayed over/under/around the basket.

The idea is that the Hebrew parallels of *over*, *under*, and *around*, can be read as trajectories through the fixed area they normally refer to. Table 3 illustrates the difference between the two types of scenarios.

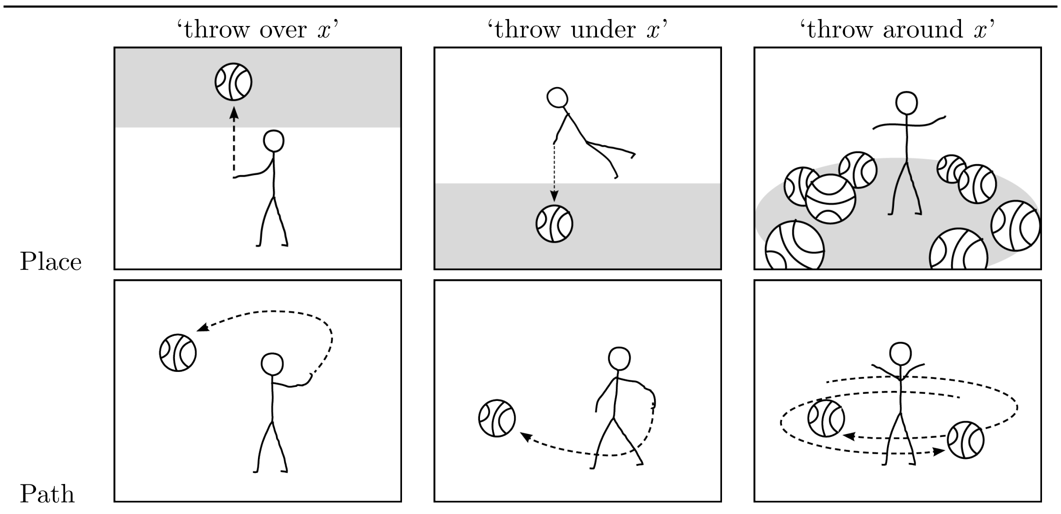


Table 3: Place and path readings of Hebrew PPs (top and bottom panels, respectively)

Note that all six scenarios include a change of location, but the prepositions capture different aspects of this change. In terms of truth conditions, the main difference is where the ball ends up: for the top scenarios, the PPs entail that the ball ends up over, under, or around the Agent, while the bottom scenarios say nothing about the ball's final location. I argue that having such pairs of distinguishable readings is a reason for non-complementarity of pronouns and anaphors in spatial PPs in addition to the logophoric option.

The main prediction of (27) for such contexts is that choosing between pronoun and anaphor will force the place or the path reading, respectively. The reverse effect should also exist, namely, contexts that strengthen one of the readings over the other will determine the type of pronominal element that the PP allows. The rest of this section shows that both sides of this prediction are supported by data from Modern Hebrew.

The first effect comes up in coordinations with a stative sentence such as 'it stayed there'. This type of continuation is natural for all the top scenarios of Table 3, but awkward with the bottom ones (since it is unclear where the ball is said to be staying). The following examples show that this test has an additional effect of making the pronouns more acceptable than the anaphors, as expected if the anaphor is licensed by the path reading of these prepositions. The oddness of anaphors in (37) follows if they coerce a path reading that is incompatible with a stative description of a final location.

(36) *Stative coordination with pronouns (ok)*:

a. wilson$_1$ zarka et ha-kadur me'al-e'a$_1$ ve-hu niš'ar šam.
Wilson threw ACC DET-ball over-3SG.F and-it stayed there
'Wilson threw the ball over her and it stayed there.'

b. wilson$_1$ zarka et ha-kadur mitaxt-e'a ve-hu niš'ar šam.
Wilson threw ACC DET-ball under-3SG.F and-it stayed there
'Wilson threw the ball under her and it stayed there.'

c. wilson$_1$ zarka et ha-kadurim misviv-a$_1$ ve-hem niš'aru šam
Wilson threw ACC DET-balls around-3SG.F and-they stayed there
'Wilson threw the balls around her and they stayed there.'

(37) *Stative coordination with anaphors (odd)*:

a. wilson$_1$ zarka et ha-kadur me'al acm-a$_1$ #ve-hu nišar šam.
Wilson threw ACC DET-ball over srefl-3SG.F and-it stayed there
'Wilson threw the ball over herself #and it stayed there.'

b. wilson$_1$ zarka et ha-kadur mitaxt le-acm-a$_1$ #ve-hu nišar šam.
Wilson threw ACC DET-ball under to-REFL-SG.F and-it stayed there
'Wilson threw the ball under herself #and it stayed there.'

c. wilson$_1$ zarka et ha-kadurim misaviv le-acm-a$_1$ #ve-hem nišaru šam
Wilson threw ACC DET-balls around to-REFL-SG.F and-they stayed there
'Wilson threw the balls around her #and they stayed there.'

The reverse effect is also attested. That is, when the context is modified to support only a path reading (38a below), the anaphor becomes obligatory. By the same token, an anaphor is banned in a context that only supports the place reading (38b).

(38) a. [kadur ha-arec]$_1$ mistovev sviv {-*o$_1$/ acm-o$_1$}.
ball.of DET-earth turns around - 3SG.M REFL-SG.M
'The earth spins around *it/itself.'

b. le-[kadur ha-arec]$_1$ yeš kim'at xamešet alafim lavjanim sviv {-o$_1$/ *acm-o$_1$}.
to- ball.of DET-earth exist almost five thousands satellites around -3SG.M REFL-3SG.M
'The earth has nearly 5,000 satellites around it/*itself.'

Another way of forcing a place reading utilizes motion verbs that select for fixed places, like *heni'ax* 'placed' or *hiciv* 'installed'. The following contexts show that only pronouns are acceptable when the prepositions *me'al* 'over' and *mitaxat* 'under' combine with such verbs.

(39) a. brajent$_1$ meni'ax et ha-kadur me'al {-av$_1$/ *acm-o$_1$}.
Bryant places ACC DET-ball over -3SG.M REFL-3SG.M
'Bryant places the ball in the air above him/*himself.'

b. brajent$_1$ meni'ax et ha-kadur mitax(a)t {-av$_1$/ *le-acm-o$_1$}.
Bryant places ACC DET-ball under -3SG.M to-REFL-3SG.M
'Bryant places the ball under him/*himself.'

A similar phenomenon is observed when adding PPs that are unambiguous path phrases. A path PP such as *la-cad ha-šeni* 'to the other side' clashes with the fixed reading of 'over' and 'under' since the ball cannot stay over a player and reach the other side of the court at the same time. This leaves the path readings of these prepositions more accessible and has the now expected result seen in (40), of anaphors being preferred for coreference with the subject.

(40) a. brajent$_1$ zorek et ha-kadur me'al {-*av$_1$/ acm-o$_1$} la-cad ha-šeni
Bryant throws ACC DET-ball above- 3SG.M REFL-3SG.M to.DET-side DET-second
šel ha-migraš.
of DET-court
'Bryant throws the ball above *him/himself to the other side of the court.'

b. brajent$_1$ zorek et ha-kadur mitax(a)t {-*av$_1$/ le-acm-o$_1$} la-cad
Bryant throws ACC DET-ball under- 3SG.M to-REFL-3SG.M to.DET-side
ha-šeni šel ha-migraš.
DET-second of DET-court
'Bryant throws the ball under *him/himself to the other side of the court.'

Again, this shows that Hebrew PPs only allow a choice between a pronoun and an anaphor to the extent that there is a choice between a place and a path reading, which in turn confirms the generalization of the overt place effect.

2.4 Paths without a place: the semantic argument

This section shows that the semantic entailments of path and place PPs also indicate that path PPs may be constructed without a place projection. Specifically, I will examine the extent to which the semantics of path and place phrases that occurs with motion verbs includes the main import of small clauses according to Hoekstra (1988), which is a result state. Hoekstra argued that constructing a small clause turns simple action verbs into causative constructions by associating them with an outcome. To illustrate, in the classic example in (41), analyzing *the metal* as the subject of *flat* turns a potentially inconsequential hammering action into a process with a defined result of the metal becoming flat.

(41) a. She hammered the metal.

b. She hammered the metal flat.

Place prepositions have a similar capacity to add a result state that is the arrival of an entity at a place. For instance, in (42), the prepositions *in*, *on*, and *next to* entail that keys should end up in, on, or next to the specified location, respectively.

(42) Mary threw her keys {in/on/next to} the cabinet.
(⊨ keys ended up in/on/next to cabinet)

Inserting path prepositions in the same contexts makes no result entailment and instead describes the keys' location before, during, or at a future stage of the action (*toward*).

(43) Mary threw her keys {to/from/through/toward} the cabinet.

These differences are clearly related to the individual meaning of each preposition. Yet, I argue that paths are also categorically prevented from specifying arrival, which is not expected if they were embedding places. My test case will be the path preposition *to*, which is the most likely to encode arrival since it selects a DP that literally names the final location of the action.

The literature on spatial PPs traditionally links *to* with the meaning of arrival at a destination (Jackendoff, 1973, 1987; Piñón, 1993; Smith, 1997), but more recent literature refines this conclusion, showing that the state of arrival is not hardwired to the meaning of *to*, but in fact can be rolled back with some supporting context. For instance, Rappaport Hovav (2008) showed that *to* phrases allow to negate arrival in non-homomorphic events (in the sense of Krifka 1989), namely events in which the course of motion does not map onto time intervals in the action that causes it, as seen in (44).[7] The following examples in (45) show that such continuations are impossible with fixed place prepositions such as *next to* or *over*.

(44) a. I threw the ball to Mary (but aimed badly and she didn't catch it).
b. We launched the rocket to the moon (but it blew up before it got there).
(Rappaport Hovav 2008 p.29)

[7] Assuming that the preposition *from* denotes the reverse path of *to* – as I do here – leads to the prediction that it would lack a result state in a similar way. That is, a 'from' PP should not entail that the discussed object left a stated location. Contexts that show this are scarce even more than parallel 'to' examples since 'from' defines a very broad goal (anywhere but the initial location), which moving objects reach at an instant. The following sentence shows it is nonetheless possible to retract *from*'s arrival entailments if the initial location is one with a threshold that takes time to leave.

label=() We launched a rocket from the moon (but it blew up before leaving the lunar gravity field.)

(45) a. I threw the ball next to Mary (#but aimed badly so it didn't get there).

b. We launched a rocket over the moon (#but it blew up before it got there).

This is supported by Bruening's (2018) observation that depictive predicates that cooccur with path phrases modify the direct object during the motion event and not the moment of arrival. To illustrate, consider an adjective like *wet* as a secondary predicate over the motion verb *walk*.

(46) a. Albert walked to the flat wet but was dry upon arrival.

b. Albert walked into the flat wet #but was dry upon arrival.

(adapted from Bruening 2018: 13)

Combining *walked to the flat* with the adjective *wet* entails that the subject was wet on their journey, but not necessarily at the point of entrance. Continuations that negate wetness at the time of arrival demonstrate this point in (46a) and are contradictory with a matched PP containing a place head (*into the flat*, 46b).

Experimental evidence from Martin et al. (2021) goes in the same direction, showing that arrival inferences are generally cancelable for English *to*, German *zu* and French *à*. The minimal pairs below show that *to* allows cancellation of arrival inferences where place prepositions such as *in*, *next to*, and *over* do not. The same effect turns up for the Hebrew prepositions *le-* 'to' vs. *be-* 'in' in (48)

(47) a. She kicked the ball **to** his face (but he dodged it).

b. She kicked the ball **in** his face (#but he dodged it).

(48) a. zarakti et ha-tik **la**-xacer aval hu lo hegi'a le-šam.
threw.1SG ACC DET-bag to.DET-yard but it NEG arrive to-there
'I threw the bag to the backyard but it didn't get there.'

b. zarakti et ha-tik **ba**-xacer #aval hu lo hegi'a le-šam.
threw.1SG ACC DET-bag in.DET-yard but it NEG arrive to-there
'I threw the bag into the yard but it didn't get there.'

Now consider path prepositions that embed place phrases overtly.

(49) a. The submarine descended to below the surface.[8]

[8]Combinations of place prepositions with *to* are rather limited in English but more widespread in Hebrew, which features phrases like *el me'ever* 'beyond' (lit. 'to beyond'), *el mitaxat* 'under' (lit. 'to under'), and so on.

b. It came from behind the dryer.

These sentences have result locations that cannot be canceled without making the sentence contradictory, as seen in (50). I conclude that paths entail results when they overtly embed places, which are clearly the source of this result meaning.

(50) a. The submarine descended to below the surface #but never got there.

b. It came from behind the dryer #but never left there.

Note that this is only possible with *to* and *from* (and Hebrew parallels). To illustrate, the above sentences turn out awkward with path prepositions like *toward* or *through*, although they describe scenarios that are physically plausible. The following Hebrew examples show the same contrast between *el* 'to' and *mi-* 'from' and other path prepositions in the language (52).

(51) a. ??The submarine descended toward below the surface.

b. ??It came through behind the dryer.

(52) a.

ha-colelet	jarda	{el/	*derex/	*lekivun}	mitaxat	li-fney	ha-jam
DET-submarine	descended	to	*through	*toward	below	to-face	DET-sea

'The submarine descended to/ *through/ *toward below the sea level.'

b.

ze	hegi'a	{mi-/	*derex}	me'axorej	ha-mekarer
DEM	arrived	from-/	*through	behind	DET-fridge

'It came from/*through behind the fridge.'

These examples show that *to*/*from*, and their Hebrew parallels are uniquely capable of embedding place phrases. I take this to mean that these prepositions have a different semantic type among the path category, a point I go back to in Section 3.3. The current bottom line is that only place prepositions drive result entailments in motion context. Table 4 shows that the acceptability of prepositions with stative verbs closely aligns with arrival entailments.

Two prepositions show mixed behavior – *into* and *onto* – to which I will return in Section 3.4. The following section lays out my proposal for the syntax of path and place PPs that makes the binding facts follow.

	in	*on*	*above*	*below*	*over*	*under*	*around*	*beside*	*beneath*	*past*	*at*
Stative verbs	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
result state	✓	✓	✓	✓	%	%	%	✓	✓	✓	%
	behind	*in front of*	*against*	*between*	*next to*	*off*	*along*	*among*			
Stative verbs	✓	✓	✓	✓	✓	✓	✓	✓			
result state	✓	✓	✓	✓	✓	✓	✓	✓			
	upon	*across*	*into*	*onto*	*to*	*toward*	*from*	*through*			
Stative verbs	✓	✓	*	*	*	*	*	*			
result state	✓	✓	✓	✓	*	*	*	*			

Table 4: Compatibility with stative verbs and the encoding of a result state in motion contexts in English prepositions

3 The syntax of space

To recap the previous section, I have argued that place prepositions in English and Hebrew set up opaque binding domains for pronouns and anaphors, and that path prepositions are transparent in this sense. This section proposes an explanation for this pattern, which can be summarized in that place PPs introduce intervening subjects. In what follows, I lay out the definitions I adopt with respect to syntactic locality (Section 3.1), before presenting the essence of the proposal (Section 3.2) and discussing its semantic consequences (Section 3.3). I then look into cases that deviate from the emerging generalization (*at*, *into*, *onto*, Section 3.4) and show that the contrast between path and place prepositions with respect to binding extends to adjuncts (Section 3.5).

3.1 Syntactic locality

The present analysis takes for granted the following notions. First, a binding domain has one and only one syntactic subject. This idea is shared by the classical definitions Governing Category (Chomsky, 1981) and Complete Functional Complex (Chomsky, 1986), as well as the minimalist terms Phase and Spellout Domain (Chomsky, 2001; Legate, 2003) which have been broadly adopted as the domains of interpretation for anaphors (Lee-Schoenfeld 2004; Canac-Marquis 2005; Johnson 2007; Quicoli 2008; Antonenko 2012; Charnavel and Sportiche 2016). The current proposal continues this line of analysis but keeps the minimal definition of locality in (53).

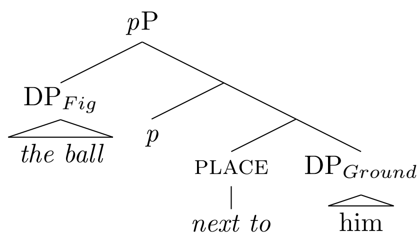
(53) The binding domain of a pronominal term x is the minimal phrase containing x and a syntactic subject.

The term “subject” refers to a hierarchically-privileged argument known historically as the external argument of V, and more recently as the specifier of a designated functional head like v or Voice (Johnson 1991; Chomsky 1995; Kratzer 1996, a.m.o). I assume that anaphors corefer with c-commanding DPs that appear as far as the nearest subject, with some language-specific variation on which DPs qualify as antecedents. Pronouns are used for coreference elsewhere, namely across intervening subjects.

3.2 Proposal: A subject in Place

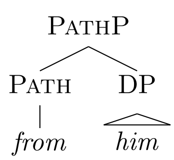
The core of the current proposal is that overt place phrases are constructed as a figure-ground relation comparable to a small clause. Paths are relations between DPs and events that may acquire the more complex figure-ground structure by embedding a whole place phrase. I specifically rely on Svenonius (2003, 2006), who posits a split syntax for spatial phrases that is analogous to the verbal domain. In this view, a functional head little p– a parallel of little v or Voice – is tasked with introducing the Figure argument.

(54) *Place phrase:*

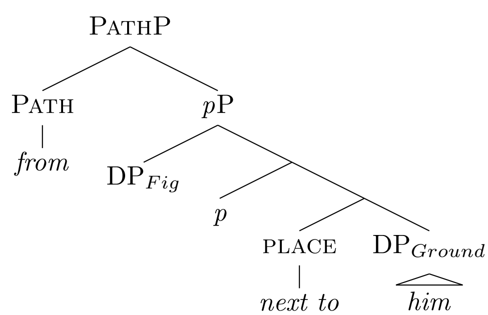


In complex PPs, a path phrase adds a layer on top of this constituent, leading to (56). The Figure's position will become irrelevant within the full sentence, since this argument will end up being pronounced as a direct object (more on this issue in the following section and footnote **??**). Paths that occur without the place projection will form the more simplistic structure in (55).

(55) *Path phrase:*



(56) *Path over place:*



This proposal keeps PATH and p as operations on fixed locations in accordance with previous analyses, but these components can occur independently of each other. When they do, paths do not require a syntactically represented Place projection but can rather manipulate a fixed location directly at the DP level, turning it to a property of motion events. A path-over-place analysis is preserved to PPs that overtly show the two layers on top of each other. The options are illustrated bellow with consequences for coreference in pronouns and entailments of arrival.[9]

	V	P	Coreference	Result
1. *Bryant*$_1$ *keeps the ball next to him*$_1$.	Stative	Place	✓	✗
2. *Bryant*$_1$ *throws the ball next to him*$_1$.	Motion	Place	✓	✓
3. *Bryant*$_1$ *takes the ball {to/from} him*$_{*1}$.	Motion	Path	✗	✗
4. *Bryant*$_1$ *takes the ball from next to him*$_1$.	Motion	Path+Place	✓	✓

Table 5: A typology of spatial PPs in English and Modern Hebrew by verb and preposition type

A small clause analysis that is limited to overt place PPs captures the acceptability of a coreferenced pronoun in lines 1, 2, 4 and not 3. This answers the question of binding into spatial PPs – at least for English and Hebrew – but opens two new questions regarding their syntax. First, it is evident that the stative construction in line 1 allows the coreferential pronoun but lacks the result entailment, suggesting that the result is not a necessary import of small clauses. Second, the range of options in Table 5 requires that path prepositions will be able to combine with elements of different types: a DP in the general case and a small clause when embedding a place phrase (lines 3 and 4, respectively). The next section shows that both issues are resolved if we assume an additional layer that encodes causation, as previously proposed in various frameworks (**?**; Wunderlich 1997; Harley 1995; **?**; Richards 2001; **?**; Folli and Harley 2007; Gehrke 2008; Ramchand 2008; Pylkkänen 2008; **?**; **?**; cf. **?**).

3.3 Getting results

I have so far highlighted the causative effects of place prepositions for the sole purpose of demonstrating the absence of place semantics in run-of-the-mill path phrases. I took for granted – following Hoekstra 1988 – that the result state is part of the import of the small clause construction. In doing

[9] I am limiting the discussion at these point to P arguments, see Section 3.5 for an extension of the proposal to nonselected PPs

so, I overlooked the lack of result entailment for place PPs appearing in stative sentences like (21), repeated below as (57).

(57) Bryant$_1$ keeps the ball next to him$_1$.

This suggests that the result state and the PP's clausal status are independent factors. The study of causation provides the needed separation in the form of a Cause head, a regular component of result meanings which is found overtly in some languages (e.g., Georgian, Khalka Mongolian; Piñón 2001; **?**). Previous studies diverge on the configuration in which Cause is plugged, and whether it is part of the lexical semantics of the verb (Dowty, 1979; **?**) or constructed in the syntax (Pylkkänen 2008; Ramchand 2008, Horvath and Siloni 2011 argue for a typology that includes both lexical and syntactic causation). For the domain of spatial PPs, there are valid reasons to link the result state with the semantics of motion verbs (Gehrke 2008; den Dikken 2010a). A classic example is the split between the verbs *jump* and *swim*, from which only the former drives a result reading with a place preposition (58).

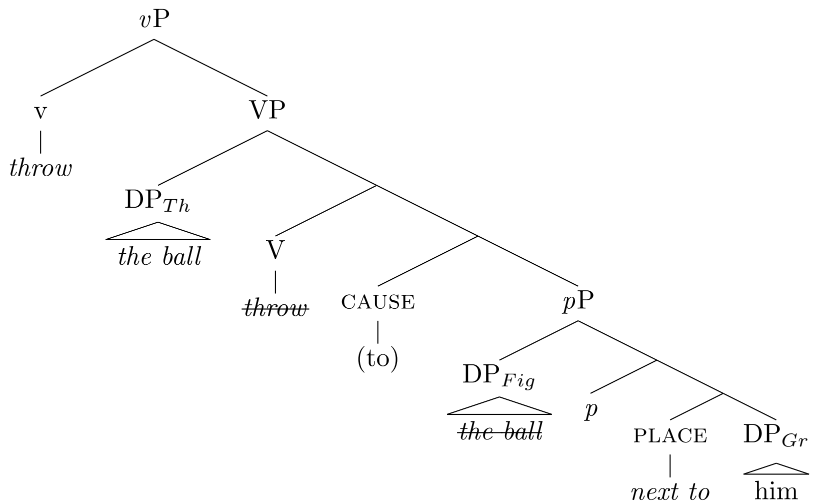
(58) a. They jumped in the lake. [✓locative, ✓result]

b. They swam in the lake. [✓locative, #result]

Yet there are also cases in which the one verb appear with and without a result. It takes the particular combination of motion verbs with place prepositions to get a consistent result entailment.[10] I will argue that, in the context of spatial PPs, Cause mediates between a motion verb and a place PP to form directional readings, and path-over-place is a transparent realization of this structure. That is, the prepositions *to*, *from*, and their Hebrew parallels that appear over place prepositions realize CAUSE rather than PATH.

[10] Gehrke (2008) discusses these readings under the term Derived Goal.

(59)



A desired consequence of this proposal is that 'to' and 'from' must have a different semantic import as part of the path-over-place configuration than as expressions of trajectories in space. The role of 'to' and 'from' here is to bring about a result state or its negation, respectively. This explains why no other path preposition in English or Hebrew can appear over place phrases, as illustrated in (60-61).

(60) Bryant moved the ball to/from/*toward/*through/*along next to him.

(61) brajent heziz et ha-kadur {el/ mi-/ *lekivun/ *derex *le-orex} mitaxt-av
Bryant moved ACC DET-ball to from toward through along under-3SG.M
'Bryant moved the ball to/from/*toward/*through/*along under him'.

In both of these examples, all English and Hebrew prepositions would be acceptable when the place prepositions are dropped. Path prepositions other than 'to' and 'from' become unacceptable when preceding a place phrase, which is expected if PATH selects a DP as a rule. Those prepositions that are acceptable over place phrases are the ones that can be recruited to encode a link to a result rather that a physical path.

These overt Cause heads should be semantically equivalent to the covert version, with a couple of privileges. One obvious difference is that overt Cause has two opposing surface options 'to' and 'from', which may encode a result that is either equal to or the negation of the state that follows. As we have seen, in the absence of these prepositions only the former meaning is available. To illustrate, in (62a), the final state of the ball can be next to Bryant or not next to him, depending on the preposition, but the version that encodes the result covertly is only understood as 'to' (62b).

(62) a. Bryant moved the ball {to/from} next to him.

b. Bryant moved the ball next to him.

Moreover, Path-over-place overcomes the well-known sensitivity of the result reading of bare place phrases to the choice of verb that was demonstrated in (58). Overt *to* or *from* enable the result reading of fixed place prepositions with *swim* verbs (63), and the same is true for the Hebrew counterparts (64).

(63) They swam {to/from} beneath the wreckage.

(64) hem saxu {el/mi-} mitaxat la-geŝer.
they swam to/from below DET-bridge
'They swam to/from below the bridge.

That is, the silent causative meaning depends on the verb in a way that path-over-place does not. I take this to mean that covert Cause is necessarily part of the VP (whether lexically or in the syntax), while the overt version may be realized in the PP, or head its own projection. The syntax trees I propose remain agnostic with respect to this detail since it does not bear on the size of the binding domain. I follow the conventional division between V and *v* for convenience with the crucial postulation that they form one maximal projection for anaphor resolution, otherwise the Theme argument should have qualified as an intervening subject.[11]

A final point to consider is the position in which the Theme of motion verb/ Figure of place preposition is pronounced (item B in the following configuration).

(65) A moved B next to C.

B's position is an open question, since (i) the causative semantics requires that these roles be assigned to the same DP; that is, the entity that undergoes motion is the one that gets to the new location; (ii) Binding restrictions on pronouns (demonstrated below) suggest that A and B are in one binding domain (below, 66a), B and C are in one binding domain (66b), and A and C are in separate domains (66c). Argument B is therefore part of two binding domains, but gets pronounced only once.

(66) a. Bryant$_1$ moved him next to the ball$_{*1}$.

[11] See (?) for thematic arguments against the *v*-V division

b. Bryant moved the ball$_1$ next to it$_{*1}$.

c. Bryant$_1$ moved the ball next to it$_1$.

A classical small-clause analysis would place B as the subject of a phrase [*B next to C*] coming at the expanse of an actual Theme argument. I will assume here that the Theme is pronounced in its canonical position, since the argument I refer to as B passes at least some diagnostics for direct objecthood.[12] My analysis avoids Figure-to-Theme raising because I have defined Theme and Figure as two thematic positions, but see Postal (1974); Lasnik and Saito (1991); Runner (2006) for arguments in favor of raising to object, and Mateu and Acedo-Matellán (2012); Bryant (2022) for a raising analysis of spatial PPs. I will instead represent the Figure argument of place PPs as a silent pronoun coreferenced with the verb's Theme. The following diagrams present the analysis for fixed places (67a), motion-place (67b), path (67c), and path over place (67d).

[12] Result place phrases divert from classic small clauses in a number of ways. First, they allow for particles to appear between the Figure DP and the place preposition, which is unexpected when they are part of the same clause.

label=() Alex put the newspaper down on the couch.

The DPs themselves pass entailment tests as if they were recipients of the action described by the verbs. Compare the place phrase in (ii) with a the canonical small clause resultatives in (iii).

lbbel=() a. Bryant throws the ball next to him (⊨Bryant throws the ball).

b. I dropped the keys in the sink. (⊨I dropped the keys.)

lcbel=() a. I want him off my ship. (⊭ I want him.) (Hoekstra, 1988)

b. She drank him under the table. (⊭ She drank him.)

Another contrast involves the acceptability of propositional anaphors standing for the result state. The following examples show that these are appropriate proxies for place phrases that appear with perception verbs – which small clause is the standard analysis for – but not when appearing with motion verbs (iv and v, respectively).

ldbel=() a. He saw a cat next to him.

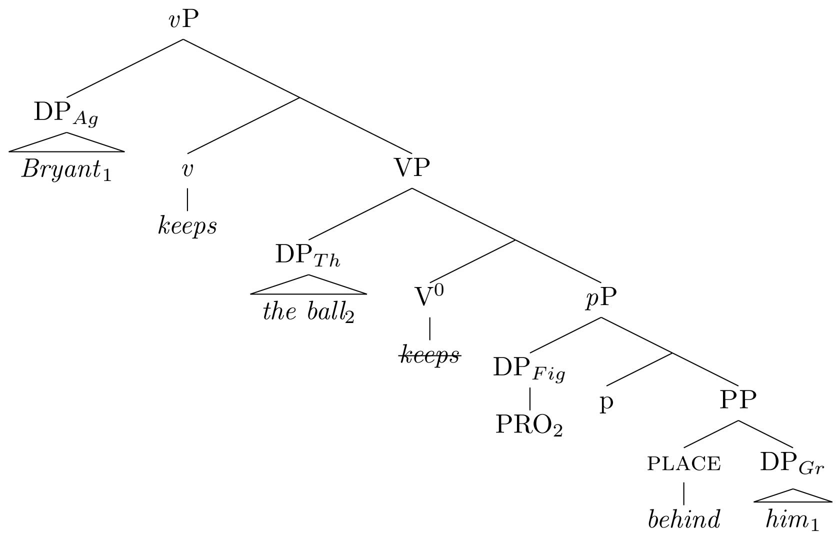
b. He saw it (it=a cat, a cat next to him)

lebel=() a. Bryant throws the ball next to him.

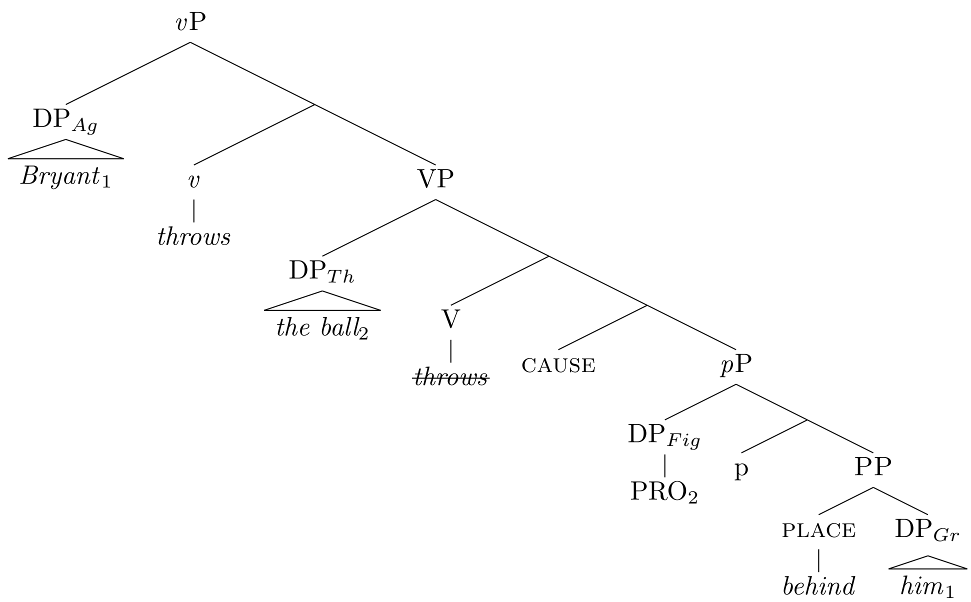
b. Bryant throws it. (it=the ball, *the ball next to him)

I take these contrasts as evidence that the syntax of result place phrases represents both the Theme and the Figure arguments.

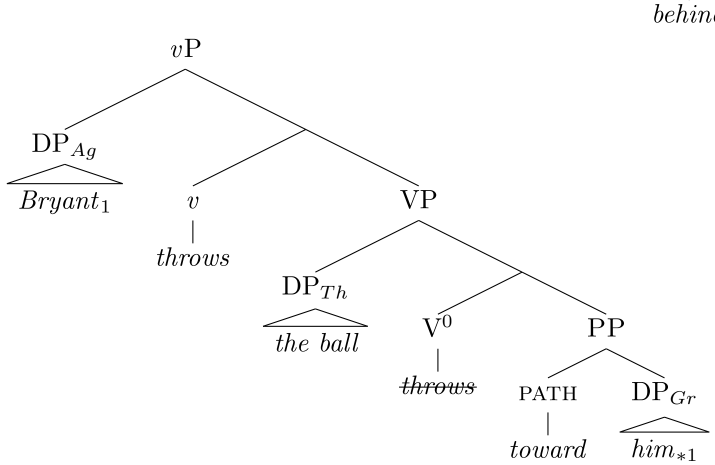
(67) a.


*v*P
DP$_{Ag}$
Bryant$_1$
v
keeps
VP
DP$_{Th}$
the ball$_2$
V^0
~~*keeps*~~
*p*P
DP$_{Fig}$
PRO$_2$
p
PP
PLACE
behind
DP$_{Gr}$
him$_1$

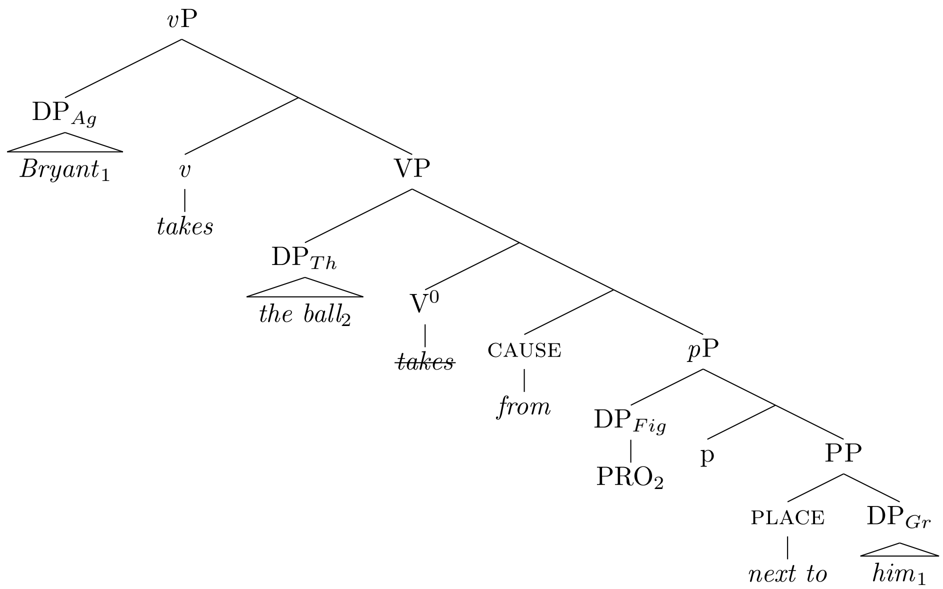
b.


*v*P
DP$_{Ag}$
Bryant$_1$
v
throws
VP
DP$_{Th}$
the ball$_2$
V
~~*throws*~~
CAUSE
*p*P
DP$_{Fig}$
PRO$_2$
p
PP
PLACE
behind
DP$_{Gr}$
him$_1$

c.


*v*P
DP$_{Ag}$
Bryant$_1$
v
throws
VP
DP$_{Th}$
the ball
V^0
~~*throws*~~
PP
PATH
toward
DP$_{Gr}$
him$_{*1}$

d.



3.4 Exceptions

I have postponed discussion of English prepositions that present an apparent exception to the overt place effect: *at*, *into*, and *onto*. My goal here is to show how they are accommodated within the proposed analysis.

The preposition *at* goes against the trends shown in Table 4 above, in the sense that it appears with stative verbs (see 68) but bans coreference in pronouns (69).

(68) They kept books **at** the daycare.

(69) They threw books **at** them$_{*1}$.

The diagnostics I introduced in Sections 2.1,2.3.1 suggest that *at* behaves like a fixed place in (68) and a path in (69). I argue that this is a correct conclusion, namely, that *at* has one reading describing the the whereabouts of an individual and another reading which can be paraphrased as *in the direction of*. We have already met prepositions that are ambiguous between dynamic and fixed location ('over', 'under', 'around'), though there was a clear link between the two readings that made it possible to derive one from the other. The ambiguity of *at* seems more like a coincidental homophony. Regardless of how it comes to be, the following examples show that the path and place readings of *at* diverge in their result entailment along the expected lines.

(70) a. The girls kept the book at the school (#but it didn't get there).

b. The fans threw some hard object at the stage (but it didn't get there).

I conclude that *at* can realize either Path or Place in a similar way to the prepositions discussed in Section 2.3.3, but with no overlap in the contexts that enable each reading.[13]

The prepositions *into* and *onto* present a different kind of exception, in which a path and a place component seem to appear together. These complex prepositions differ from run-of-the-mill path-over-place sequences, first in the order in which they appear (place before path), and then in the inability to separate the two layers by modifiers (71a below vs. 71b).

(71) a. **from** (two meters) **behind** the baseline

b. **in** (*two meters) **to** the sand

It is therefore expected hat *onto* an *into* will pass both place and path diagnostics. In reality they turn out to show mixed results, being infelicitous with stative verbs (72), entailing a result location (73), and banning coreference in pronouns (74).

(72) *They stayed {into/onto} the bus.

(73) They climbed {into/onto} the bus #but never got there.

(74) Corporal Crump$_1$ pinned the medal onto him$_{*1}$/himself$_1$. (Wechsler 1997: p.15)

In other words, *into* and *onto* pass and fail place diagnostics at the same time. I argue that this is a special case in which path prepositions are adjoined by a place particle like the following.

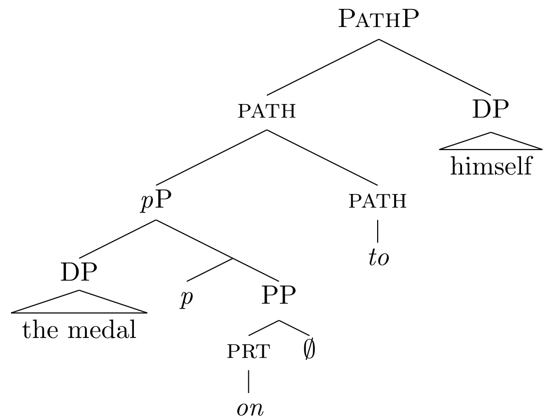
(75) a. Put the hat **on**.

b. Take the cat **in**.

Standard analyses of particles view them as intransitive small clauses (Kayne 1995; den Dikken 1995; Koopman 2000; Svenonius 2010). In the system adopted here, this makes them *p* phrases with a subject but no object. Combining this with a path phrase – which has an object but not a subject – we get a configuration in which the subject and the object are split between different maximal projections (76).

[13] The place version of *at* should allow a coreferenced pronoun. This is a prediction of the current analysis that cannot be verified since the stative *at* does not take pronominal complements in general.

(76)



The object is not part of the binding domain that contains the subject and is therefore not protected from Principle B, explaining the ban on coreference in (74). This also explains why such compound prepositions are limited in English to *on* and *in* (i.e., no *belowto*, *behindto*, etc.), as these are the main English particles in the category P.

3.5 Application to Nonselected PPs

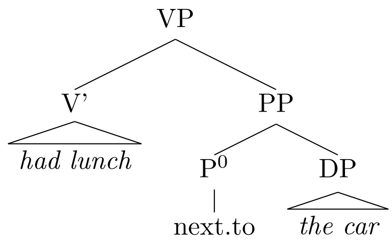
Most of this paper focused on comparing place and path prepositions that surface as selected arguments, but place prepositions have a much broader distribution in their capacity as descriptions of general location for any kind of event or state, as in (77).

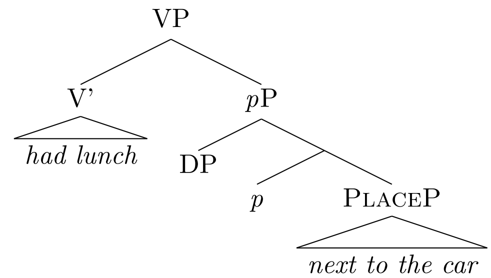
(77) The girls {stayed/talked/ran/had lunch} next to the car.

My goal in this final subsection is to show that the place-path divide withe respect to syntactic subjects also applies to nonselected PPs that are traditionally analyzed as adjuncts. This class predominately consists of place prepositions, but there is also a limited nonselected use of paths like the example in (78).

(78) They screamed {from/through} the window.

The classic analysis of nonselected PPs (e.g., Fillmore 1966) is of a simple adjunct phrase containing P and its complement, which fits path and place PPs alike (79a below). The question now is whether the more complex structure on the right (79b) that includes a small clause is a better representation of nonselected place PPs.

(79) a. 

b. 

As before, we expect two types of effects if nonselected PPs are small clauses. First, the PP should form a binding domain and allow pronouns to be coreferenced with DPs higher up. Contexts like (80) confirm that this is possible.

(80) Alex mentioned John_1 next to him_1.

This sentence has a reading in which Alex mentions John while standing next to John, which suggests that the PP is its own binding domain. Alternatively, it could be argued that the adjunct is not in *John*'s c-command domain, in which case no Principle B violation is predicted regardless of the PP's internal syntax. However, Principle C violations that come up when we switch the name and the pronoun suggest that there is c-command between these positions (Johnson, 1991).

(81) Alex mentioned him_1 next to John_{*1}.

Crucially, a nonselected path bans a coreferenced pronoun in this position. Think of the following sentence in a context where an Agent is faced with a number of sound-transmitting pipes going in different directions, one of them directed back toward the Agent.

(82) John_1 shouted toward $\{\text{him}_{*1}/\text{himself}_1\}$.

At least some English speakers accept an anaphor in this context, while a coreferenced pronoun is out for everyone. The contrast between this and (80) follows if there is a *p*P layer over the place preposition that is lacking for the path. This means that some DP is the subject of the place phrase, which should lead to an entailment that its referent is at the location named by the PP. The following sentence shows that no particular DP position has to take this role, but that one of them has to. That is, we can cancel the location entailment that is encoded by the place adjunct for any individual participant, but not for all of them at the same time.

(83) a. We heard Alex introducing the girls to each other in the hall (but {we/she/they} weren't actually there).[14]

b. We heard Alex introducing the girls to each other in the hall (#but none of us were actually there).

These sentences can describe either a situation in which we hear an introduction taking place in the hall from another location, or one where we are in the hall hearing an introduction somewhere else. A scene in which the sound of the girls being introduced is played in the hall and heard by us from another room is technically possible but is not described by (83). Now consider a similar sentence with a path preposition.

(84) We heard Alex introducing the girls to each other through the window.

The path phrase *through the window* (84) does not require that any participant go through the window. That is, it can be true if the earwitness and the introducing event are located somewhere else and only a recorded sound of the event goes through. This follows if nonselected place prepositions have a subject that is missing in paths.

4 Conclusion

The main question of this paper was what determines the choice of coreferential expressions in PPs. I proposed that the key factor is whether the preposition is understood as a fixed place or a path. I followed a split-PP approach and argued that place prepositions surface in the context of a functional head p that introduces an intervening subject, making the place phrase comparable to an embedded clause for binding considerations. I showed that fixed place prepositions are consistent not only in allowing coreferenced pronouns but also in entailing arrival at the location they mention, which does not happen with path prepositions. Complex PPs in which a path embeds a place were shown to pattern with place phrases on both counts (forming binding domains and encoding arrival rather than the trajectory taken. The necessity of the overt place component for these effects suggests that in their absence, paths can modify locations directly from bare DPs. That is, the shift from an

[14] I thank an anonymous reviewer for pointing out this type of examples.

object to the object's location on the way to a path description happens at the conceptual level and is not represented in the syntax.

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